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Unveiling the effects of multi-element interaction on phase compatibility and interfacial bonding in AgCuSnTi/diamond brazing system via first-principles calculations



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Highlights

- Multi-element interactions in the AgCuSnTi/diamond brazing system were investigated.
- Adsorption and charge-transfer features of Ag, Cu, Sn, and Ti atoms on diamond were analyzed.

- Thermodynamic stability and mechanical properties of doped brazing phases were evaluated.
- Bonding strength and fracture behavior at the diamond interface were characterized.

Abstract

Diamond brazing is a critical joining technique for advanced thermal management and cutting applications, where joint reliability primarily depends on the resulting interlayer. In this study, first-principles calculations were conducted to investigate the effects of multi-element interaction on phase compatibility and interfacial adhesion in the AgCuSnTi/diamond brazing system. Adsorption and charge-transfer characteristics of Ag, Cu, Sn, and Ti atoms on diamond were first examined, revealing that Ti exhibited the strongest electronic interaction and acted as the primary wetting agent that enhanced interfacial bonding. To capture the compositional complexity of the interlayer, major phases were doped with additional filler elements. The calculated formation energies and mechanical parameters demonstrated strong composition dependence, with certain doped configurations compromising thermodynamic and mechanical stability. Moreover, layer-specific segregation at the diamond interface indicated that interface-closest segregation severely weakened bonding strength and facilitated premature fracture. Electronic structure and charge density analyses clarified the bonding disruption mechanisms. These results could advance the understanding of microstructure-property relationships in carbon-based materials brazed with AgCuSnTi fillers.



Keywords

Diamond brazing; First-principles calculation; Ag-based filler alloy; Reaction phase; Bonding strength

1. Introduction

As a strategic material for advanced engineering fields, synthetic diamond has attracted considerable attention due to its exceptional physical properties, including wide bandgap, high thermal conductivity, ultralow dielectric loss, broad optical transmission range, and extreme hardness [1,2]. These outstanding properties stem from perfect sp^3 -hybridized carbon structures and robust covalent bonding networks [3]. To fully realize its multifunctional potential, diamond must be reliably joined to metals, where metals provide mechanical support and heat transfer, while diamond contributes superior thermal transport and electromagnetic properties [4]. Vacuum brazing has emerged as an effective technique for bonding carbon-based materials or ceramics to themselves or metals, offering high assembly precision, excellent strength, and minimal thermal damage [5,6]. Notable applications include diamond optical windows, where large-diameter diamond disks are brazed to copper rings for use in fusion reactors [7,8]. In another advanced application, steel-brazed diamond abrasives enable ultra-precision grinding operations [9]. Additionally, diamond-reinforced Al and Cu matrix composites, as well as diamond Al- or Cu-clad laminates, are emerging as advanced thermal management solutions for next-generation electronics [10,11]. All of these cutting-edge applications critically rely on diamond brazing technology.

Current diamond brazing employs interlayers including Ag-based, Cu-based, and Ni-based multi-component filler alloys [12,13]. Ni-based alloys offer superior strength at elevated temperature ($> 900^\circ\text{C}$), but the presence of catalytic element Ni and the required high brazing temperature tend to promote diamond graphitization and microcrack initiation [14,15]. Cu-based fillers, though cost-effective, suffer from poor wettability and oxidation susceptibility [16]. In comparison, Ag-based (particularly AgCu) fillers, such as the widely used eutectic composition 72Ag-28Cu (wt.%), demonstrate balanced advantages with a moderate liquidus temperature (780°C), good fluidity, and compatible thermal expansion ($18.5 \times 10^{-6} \text{K}^{-1}$), achieving reliable joint strength [17]. Additionally, a ternary Ag-Cu- x interlayer, where x represents a melting-point depressant (e.g. Sn, In, or Si), has been developed to further reduce processing temperatures. Among these, Sn is particularly advantageous, effectively lowering the eutectic temperature while maintaining mechanical performance, outperforming In-induced embrittlement and Si-induced viscosity increases [18,19]. To address the chemical inertness of diamond, active elements such as Ti, Cr, or V are incorporated into the filler. Notably, Ti has been identified as the most effective: as little as 0.5wt.% Ti can reduce the contact angle between diamond and Ag from 120° to 45° at 1000°C [20]. Ti reacts in situ with diamond to form a stable TiC interfacial layer, while its solubility in AgCu ensures a homogeneous distribution without significantly impairing the ductility of the filler matrix [21]. These synergistic effects make the quaternary Ag-Cu-Sn-Ti filler system a promising solution for brazing carbon-based hard materials. Thanks to extensive studies conducted by our group, both filler composition (AgCu- m Sn- n Ti, $m=5, 10, 15, 20$, $n=1, 2, 3$)

and brazing process parameters (brazing temperature 720–780°C, holding time 5–20 min) have been systematically optimized to achieve superior interfacial microstructures and mechanical performance [7,22]. It was prepared by ball milling 72Ag–28Cu eutectic alloy powders, Sn powders, and TiH₂ powders. The optimized AgCu–10Sn–1Ti filler, corresponding approximately to an atomic ratio of Ag₂₇Cu₁₈Sn₄Ti₁ (at.%), under brazing conditions of 750°C for 10 min, was confirmed to produce joints with shear strength up to 256 MPa and stable microstructures [22]. Furthermore, the AgCuSnTi filler has shown compatibility with C/C composites and graphite [23,24].

Processing parameters and compositional adjustments promote the formation of diverse compounds at bonding interfaces, exhibiting distinct thermal/mechanical properties [25]. Ag, as a precious metal, exhibits relatively low chemical reactivity and does not readily react with other elements during brazing. As a result, it tends not to form stable Ag-based intermetallic compounds (IMCs) and typically remains as a matrix component without actively participating in the formation of extensive interfacial products. In contrast, Cu shows much higher chemical activity, often exhibiting a higher diffusion rate and stronger reactivity during brazing. Cu readily reacts with elements in the base material, such as Ni, Ti, or other transition metals at the interface, leading to the formation of Cu-based IMCs. Our early studies [7] revealed that the reaction layer in diamond/AgCuSnTi joints primarily comprises Ag and Cu solid solutions (Ag(s,s)+Cu(s,s)) with different doping concentrations. In addition to being captured by carbon atoms to form the TiC layer, Ti also promotes the formation of CuSn₃Ti₅ IMCs. Under different processing conditions, Sn₃Ti₅ and CuTi phases were likewise detected within the reaction layer. In ceramic/metal brazed joints, the interfacial transition between ceramic substrates and filler alloys presents critical challenges due to the mismatch in atomic bonding characteristics and coefficient of thermal expansion (CTE), which dictates stress distribution profiles. These mismatches strongly influence residual stress distributions and often promote crack initiation within the reaction layer under mechanical loading. However, the mechanical consequences of elemental doping on the brazing-phase constituents remain quantified insufficiently, despite their critical importance for guiding microstructural optimization. Moreover, current limitations in spatial and temporal resolution hinder the precise evaluation of filler element segregation at the diamond interface. Identifying vulnerable interfacial regions remains essential for understanding the fracture behavior and long-term reliability of brazed joints.

Computational simulations serve as powerful tools for the description and understanding of atomic-scale experimental phenomena. First-principles calculations based on density functional theory (DFT) have proven effective in predicting solute segregation behavior and interfacial bonding characteristics [26,27]. J.L. Wang et al. [28] demonstrated that Al, Mn, and Ni tend to segregate to coherent Cu/Fe interfaces, reducing interfacial energy and slowing Cu precipitate coarsening. P. Liu et al. [29] reported that N and B doping at graphene/Al interfaces weakens C–C bonds but enhances Al–C bonding, thereby improving

interfacial integrity. Similarly, X.K. Chen et al. [30] revealed the atomic-scale origins of improved strength and corrosion resistance in Au-doped TA9/Sn-6Zr/ZrO₂ joints by analyzing surface properties and mechanical parameters. Collectively, these studies underscore the capability of first-principles methods to predict solute behavior, segregation energetics, and their synergistic effects on interfacial stability and strength.

This study explored the effects of multi-element interaction on phase compatibility and interfacial properties in the diamond/AgCuSnTi brazing system. The analysis began with evaluating the adsorption behavior and charge-transfer kinetics of individual filler elements on the diamond surface. Subsequently, doped brazing-phase models were constructed to assess their thermodynamic stability and mechanical properties, with a focus on their implications for interlayer strength. Finally, representative diamond-side heterointerface models incorporating elemental segregation were developed to examine their effects on interfacial bonding and adhesion. This multiscale investigation (spanning atomic-level adsorption to reaction-phase stability, and interfacial structure) provides a mechanistic link between experimental observation and computational prediction, informing elemental compatibility and interfacial weaknesses in the diamond/AgCuSnTi brazing system.

2. Computational methods

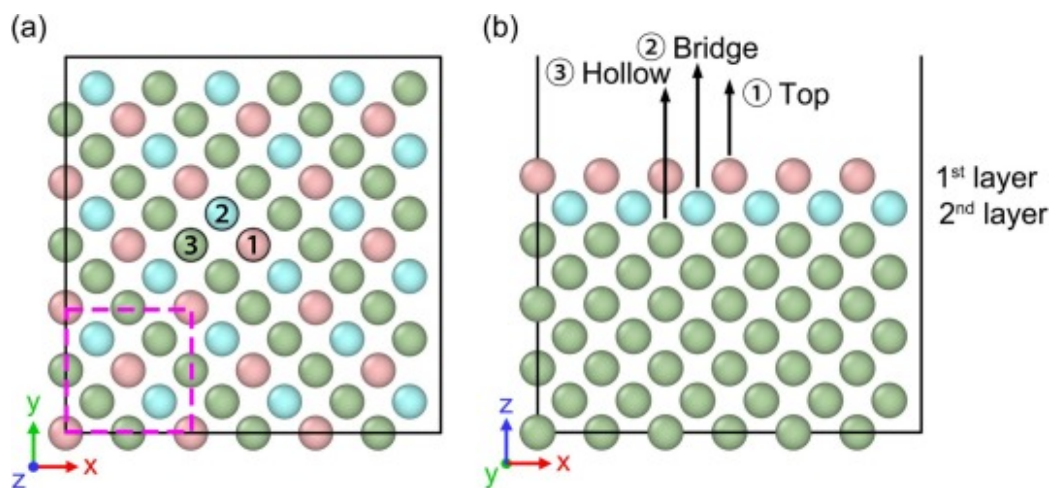
First-principles calculations were performed using the Vienna Ab initio Simulation Package (VASP), with ion-electron interactions described by the projector augmented wave (PAW) method [31,32]. The exchange-correlation energy was treated using the Perdew-Burke-Ernzerhof (PBE) functional within the generalized gradient approximation (GGA) framework [33,34]. Convergence tests established optimal computational parameters: A plane-wave energy cutoff of 450 eV, and total energy and atomic force converged within 10^{-6} and 0.02 eV/Å, respectively. Brillouin zone sampling was conducted using a Γ -centered Monkhorst-Pack k-point mesh with a resolution of $0.06\pi \text{ \AA}^{-1}$ [35]. A vacuum layer of 15 Å was introduced to eliminate periodic interactions in slab models, and the bottom four atomic layers were fixed during structural relaxation. Post-processing analyses were partially carried out using the VASPKIT toolkit [36].

In this study, the diamond(001) crystallographic plane was strategically selected as the model surface due to its broad practical applications [37,38], as well as higher chemical reactivity compared to the (111) plane [[39], [40], [41]]. Unless otherwise specified, the term “diamond surface” hereafter refers to the (001) plane. To ensure accurate representation of both surface and bulk-like properties, a sufficiently thick slab model is essential. Accordingly, a nine-layer diamond slab was adopted, consistent with previous studies demonstrating that this thickness yields well-converged surface energies and structural features [42].

3. Results and discussion

3.1. Adsorption behaviors of Ag, Cu, Sn, and Ti filler elements on diamond(001)

To gain fundamental insights into the early stages of atomic contact and wetting during the brazing process, it is essential to understand the preferential adsorption behavior of individual filler elements (Ag, Cu, Sn, and Ti) on diamond surfaces. Such knowledge also aids in predicting phase growth modes and elemental distribution at the interface. Considering surface size effects, a 3×3 supercell of the nine-layer diamond(001) slab ($x//[100]$, $y//[010]$, and $z//[001]$) was selected as the representative substrate. Based on symmetry analysis, three distinct adsorption sites were identified, as illustrated in Fig. 1. Notably, the bridge site allows for two possible configurations, however, only the configuration with a carbon atom directly beneath the adsorbate was considered, as it is expected to exhibit stronger adsorption due to enhanced local interaction.



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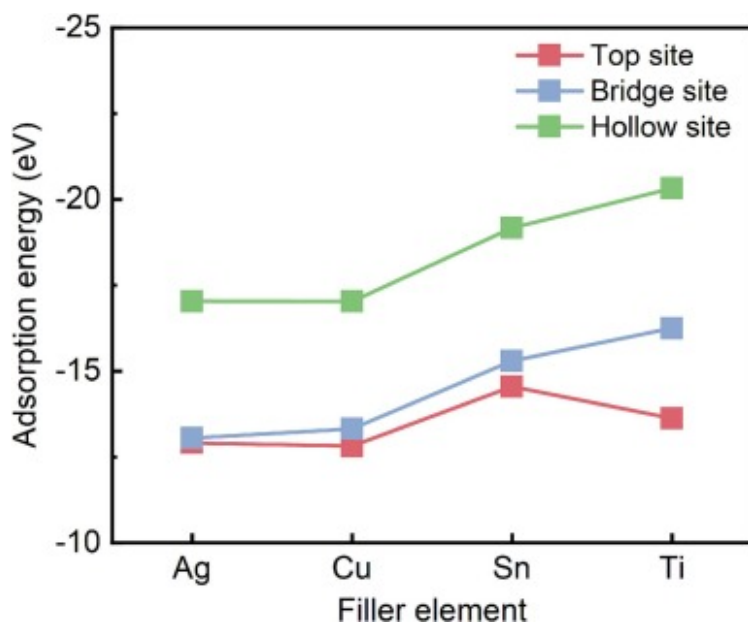
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Fig. 1. Schematic diagrams showing the representative adsorption sites on a 3×3 diamond(001) surface model with 9 atomic layers: a) Top view and b) side view. The pink dashed square in a) represents the minimal periodic conventional unit cell. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The adsorption energy (E_{ads}) quantifies the thermodynamic favorability of adsorption:

$$E_{\text{ads}} = E_{\text{substrate-adsorbate}} - E_{\text{substrate}} - \mu_{\text{adsorbate}} \quad (1)$$

It is defined as the energy difference between the substrate-adsorbate system ($E_{\text{substrate-adsorbate}}$) and the sum of the clean substrate ($E_{\text{substrate}}$) and the isolated adsorbate ($\mu_{\text{adsorbate}}$). Here, the chemical potential ($\mu_{\text{adsorbate}}$) refers to the energy per atom of each filler element in its most stable bulk phase (FCC-Ag, FCC-Cu, β -Sn, and HCP-Ti), reflecting the eutectic nature of the brazing alloy and the single-atom adsorption configurations. A negative E_{ads} indicates an exothermic process, with more negative values corresponding to stronger binding between the adsorbate and the substrate, and thus a more thermodynamically favorable and stable adsorption configuration. As shown in Fig. 2, the hollow site consistently yields the lowest E_{ads} for all investigated elements (Ag, Cu, Sn, and Ti) on diamond, identifying it as the most energetically favorable site. This preference can be attributed to the hollow site offering a higher coordination environment, which enhances bonding interactions between the adsorbed atom and neighboring surface carbon atoms, thereby stabilizing the adsorption structure.



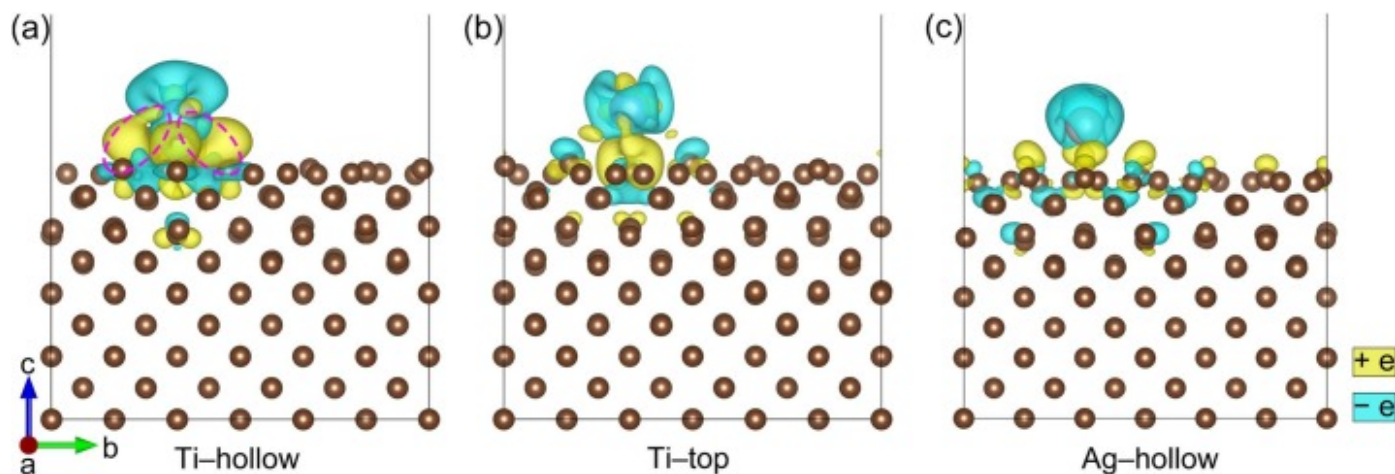
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Fig. 2. Adsorption energies of filler elements at different adsorption sites on the diamond(001) surface.

Among the investigated elements, Ti and Sn exhibit markedly stronger adsorption on the diamond surface compared to Ag and Cu, with Ti consistently displaying the most negative values. This behavior aligns with Ti's well-known carbide-forming tendency, as its partially filled 3*d* orbitals effectively hybridize with the 2*p* orbitals of surface carbon atoms, resulting in the formation of strong covalent bonds. Such robust interactions underpin Ti's established role as an active element in enhancing wetting and interfacial bonding in diamond brazing systems. The distinct adsorption capabilities of Ag, Cu, Sn, and Ti lead to differences in structural stability at the interface. Over time, thermodynamically more stable atoms preferentially occupy surface sites, suggesting that Ti tends to accumulate at the diamond interface, thereby promoting the formation of a continuous TiC reaction layer. For context, literature values place Ag and Cu as weakly bound on graphitic carbon (absolute magnitudes 0.1–0.9 eV for Cu and 0.19–0.70 for Ag), whereas Ti exhibits much stronger chemisorption (2.2–2.4 eV) [43,44]. On diamond surfaces, both calculations and experiments indicate far stronger titanium–carbon interaction: reported adsorption/adhesion energies on oxidized diamond are on the order of ~7 eV for Ti, and first-principles studies on diamond(111) find the Ti>Sn>Cu with Ti magnitudes of ~9 eV, being almost twice that of the others [45,46].

To understand the adsorption preference and stability, three representative configurations were selected for detailed analysis: Ti adsorbed at the hollow site (Ti–hollow), Ti at the top site (Ti–top), and Ag at the hollow site (Ag–hollow). The comparison between Ti–hollow and Ti–top focuses on the differential charge density maps, which directly correspond to the first observed trend, that the hollow site consistently yields the most favorable adsorption energy for all filler elements. This analysis provides microscopic insight into why the hollow site is energetically preferred. By contrast, the comparison between Ti–hollow and Ag–hollow is linked to the second observed trend, that Ti exhibits markedly stronger adsorption than Ag and Cu. This analysis highlights how charge transfer behavior differs between Ti and Ag, which is particularly important because Ag is the predominant constituent in the conventional 72Ag–28Cu brazing alloy. Together, these analyses clarify both the origin of site preference and the element-specific role of filler components in interfacial bonding. The charge density difference maps provide a visual representation of electron redistribution upon adsorption. Regions of electron accumulation (yellow) and depletion (blue) indicate the extent and direction of charge transfer between the filler atom and the surface carbon atoms. As shown in Fig. 3, the bottom six carbon layers remain largely unaffected, exhibiting negligible charge redistribution, which suggests that the bulk structure of the diamond substrate is preserved during adsorption. In contrast, significant structural rearrangement is observed in the topmost three carbon layers, particularly in the vicinity of the adsorbed atoms, indicative of new chemical bond formation. This localized bonding behavior confirms that charge transfer is primarily confined to the near-surface region around the adsorption site.



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Fig. 3. Charge density differences for filler elements adsorbed on the diamond(001) surface at different adsorption sites: a) Ti-hollow, b) Ti-top, and c) Ag-hollow. Yellow and blue regions represent electron accumulation and depletion, respectively. The isosurface value is uniformly set to 0.005 e/Bohr^3 . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Notably, the Ti-hollow configuration (Fig. 3a) exhibits a more extensive charge redistribution region compared to the Ti-top configuration (Fig. 3b), characterized by pronounced electron depletion around the Ti atom and accumulation near adjacent surface carbon atoms. In contrast, the Ti-top configuration displays a markedly weaker and more localized charge redistribution, indicative of a less favorable bonding environment. The enhanced interaction in the hollow site arises from its favorable geometric coordination, where the Ti atom is symmetrically surrounded by four surface carbon atoms in a planar configuration (Fig. 1a). This arrangement enables optimal spatial overlap between the partially filled Ti 3d orbitals and the 2p orbitals of neighboring carbon atoms, facilitating the formation of multiple directional covalent bonds. Such multi-center orbital hybridization enhances charge transfer and contributes to the thermodynamic stability of the adsorption structure, thereby explaining why the Ti-hollow site yields the lowest adsorption energy among all configurations. Additionally, the equilibrium interfacial distances (d) between the adsorbed atom and the diamond surface were analyzed for the relaxed structures. The Ti-

hollow configuration exhibits a shorter bond length of 1.478 Å, compared to 1.773 Å in the Ti-top configuration, further reflecting the stronger chemical interaction and more stable bonding at the hollow site.

In comparison, the Ag-hollow configuration (Fig. 3c), exhibits much weaker electron redistribution, which is consistent with its relatively low adsorption energy. From an electronic structure standpoint, Ag possesses a filled $4d^{10}5s^1$ configuration, which limits its ability to participate in orbital hybridization or donate charge, resulting in a weak interaction with the diamond surface. Overall, the spatial extent and intensity of interfacial charge transfer observed in the charge density maps are in good agreement with the computed adsorption energies, confirming the validity of the adsorption models. During interfacial reactions involving AgCuSnTi filler alloys, Ti atoms are expected to preferentially bind to the diamond surface due to their high adsorption energy and strong bonding capacity. This leads to the formation of a reactive wetting layer that promotes interfacial adhesion. Sn, which exhibits intermediate adsorption strength, may contribute to moderate chemical bonding and assist in modifying the surface chemistry or enhancing filler spreading. In contrast, Ag and Cu interact only weakly with the diamond substrate and are therefore more likely to remain within the bulk of the filler material, serving primarily as the metallic matrix. This element-specific interfacial behavior determines the vertical compositional gradient and governs the microstructural evolution near the diamond/filler interface, which ultimately plays a critical role in joint strength and reliability.

3.2. Filler element doping and its influence on the properties of brazing phases in the diamond/AgCuSnTi system

3.2.1. Doping preference and phase stability

This section examines the key brazing phases formed in the diamond/AgCuSnTi reaction system, including the TiC reaction layer at the diamond interface, metallic matrix phases (Ag and Cu), and IMCs CuTi and CuSn₃Ti₅. During solidification, these phases inevitably incorporate other filler elements, often forming substitutional solid solutions due to the relatively large atomic radii of alloying atoms. As a result, doping-induced variations in phase stability and mechanical behavior merit detailed investigation, with the broader objective of understanding the interactions among multiple filler elements. In metallic solid solutions, typical doping concentrations range from a few atomic percent up to approximately 10 at.%, depending on solubility limits and phase equilibria. Two representative doping concentrations were considered, approximately 3% and 6%, corresponding to the substitution of one or two atoms in the simulation cell. These values fall within experimentally observed concentration ranges and are appropriate for capturing the features of doping effects.

Table 1 summarizes the calculated crystallographic parameters of the key brazing phases, along with the details of the supercell models used for doping simulations. To represent disordered atomic distributions, doped models with nearly equal dimensions along the x , y , and z directions were constructed using the MCSQS module within the ATAT package [47,48]. This code generates special quasirandom structures by optimizing atomic correlation functions to closely approximate those of an ideal random alloy. All constructed random solid-solution models achieved objective function values below -0.92 , indicating excellent agreement with the target correlation functions and confirming that the atomic arrangements provide a reliable representation of ideal disorder. The doping strategy involved substituting filler elements into host lattice sites while excluding the intrinsic metallic atoms of each phase. For multicomponent phases, configurations involving the simultaneous substitution of two atoms were also considered to reflect more complex compositional scenarios. To ensure clarity in model identification, a systematic naming convention was adopted in the format “*dopant(host)-n*”, where “*dopant*” denotes the substituted filler element, “*host*” indicates the atom or atoms being replaced, and “*n*” refers to the number of substitutions. For single-metal phases such as TiC, Ag, and Cu, the host species is omitted for simplicity. In contrast, for IMCs containing multiple metallic constituents, co-substitution of two different host atoms was also considered. For instance, in the doped CuSn_3Ti_5 system, the label $\text{Ag}(\text{Cu},\text{Sn})-2$ refers to a configuration in which one Cu atom and one Sn atom are simultaneously replaced by Ag atoms.

Table 1. Crystallographic information of brazing phases in the diamond/AgCuSnTi system and model details for doping simulations, compared with experimental or theoretical lattice constants.

Phase	Space group	Lattice constants (Å)	Supercell	Number of metal atoms
TiC	Fm-3m	$a=b=c=4.332$ $a=b=c=4.329$ [49]	$2 \times 2 \times 2$	32
CuTi	P4/nmm	$a=b=3.120$, $c=5.918$ $a=b=3.122$, $c=5.924$ [50]	$2 \times 2 \times 2$	32
CuSn_3Ti_5	$P6_3/\text{mcm}$	$a=b=8.213$, $c=5.557$ $\alpha=\beta=90^\circ$, $\gamma=120^\circ$ $a=b=8.162$, $c=5.574$ $\alpha=\beta=90^\circ$, $\gamma=120^\circ$ [51]	$1 \times 1 \times 2$	36

Phase	Space group	Lattice constants (Å)	Supercell	Number of metal atoms
Ag	Fm-3m	$a=b=c=4.146$	$2\times 2\times 2$	32
		$a=b=c=4.090$ [52]		
Cu	Fm-3m	$a=b=c=3.628$	$2\times 2\times 2$	32
		$a=b=c=3.608$ [53]		

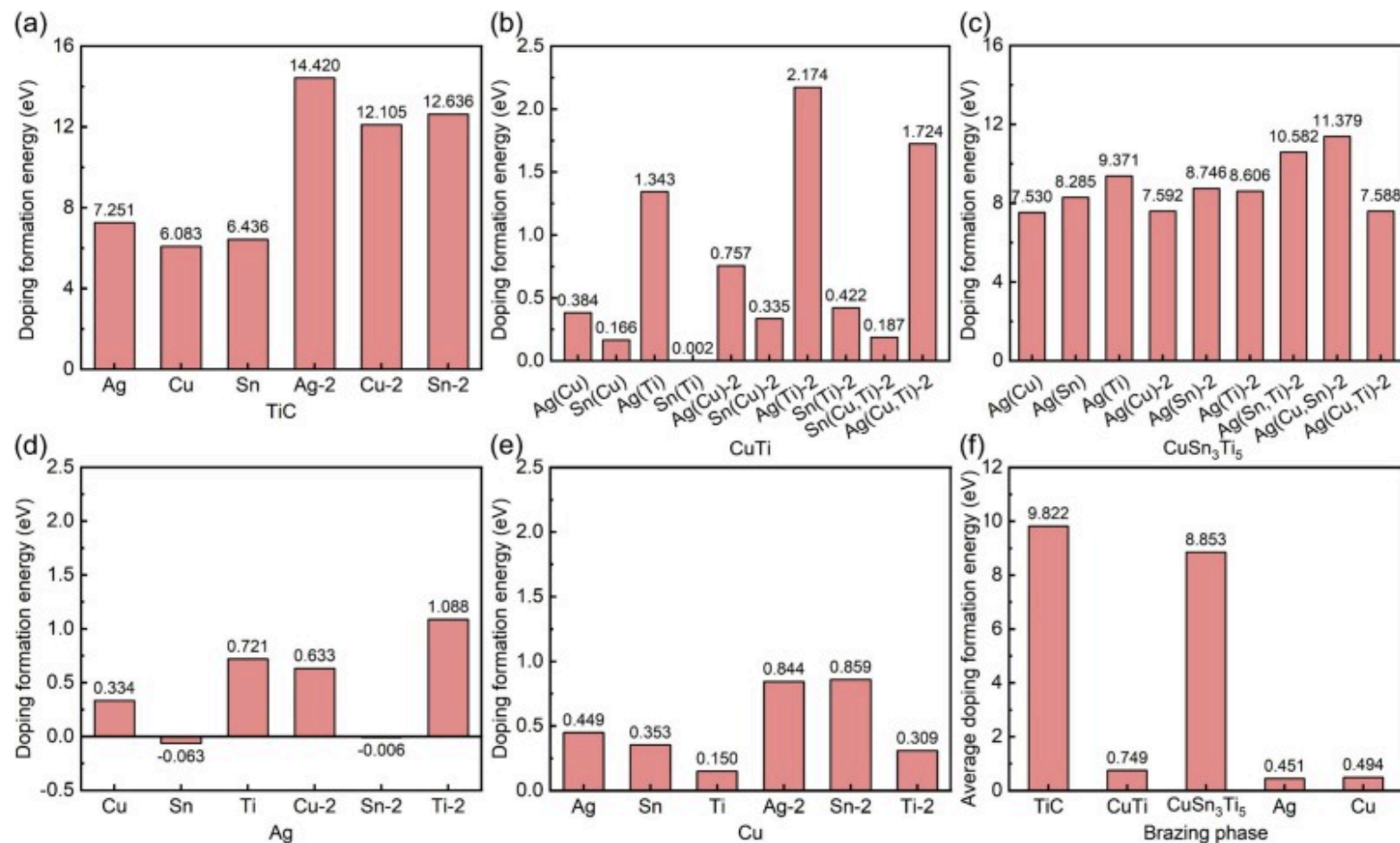
Following structural relaxation, it was observed that many cubic phases transformed into orthorhombic structures upon doping, indicating a reduction in crystal symmetry. This symmetry breaking arises primarily from atomic size mismatch, localized lattice strain, and charge redistribution introduced by the dopant atoms. Consequently, the system relaxes into a lower-symmetry configuration that is energetically more favorable under the given doping conditions. Doping formation energy (E_{dop}) is a key thermodynamic metric to evaluate the stability of dopant incorporation into host phases, informing structural tolerance and compositional adaptability of different brazing phases under multi-element interactions:

$$E_{\text{dop}} = E_{\text{doped}} - E_{\text{pure}} + \sum_{i=1}^n n_i \mu_i^{\text{host}} - \sum_{j=1}^n n_j \mu_j^{\text{dopant}} \quad (2)$$

where E_{doped} is the total energy of the doped system, E_{pure} is the total energy of the pure (undoped) supercell. μ^{host} and μ^{dopant} represent the chemical potentials of the replaced host atom and the introduced dopant atom, respectively. A negative E_{dop} indicates that the doping process is thermodynamically favorable, meaning the doped configuration is more stable than the pure phase under the given chemical potentials. A positive E_{dop} indicates that the incorporation of dopant atoms is not thermodynamically spontaneous and therefore requires external driving forces, such as elevated temperature or pressure. This is a normal condition in brazing, where the high-temperature environment and local compositional fluctuations provide sufficient kinetic energy to overcome such energetic barriers, thereby enabling the formation of complex reaction phases and dopant-stabilized structures in the interlayer.

Fig. 4a–e shows the E_{dop} for doped phases with different filler elements and doping concentrations. Previous studies provide useful benchmarks for metallic systems. For instance, V.I. Razumovskiy et al. [54] performed DFT calculations for solute segregation at the Cu $\Sigma 5(210)$ grain boundary and compared them with diffusion experiments, reporting an Ag segregation

enthalpy of about 39–40 kJ/mol (≈ 0.4 eV), with excellent agreement between theory and experiment. Similarly, A.Y. Lozovoi et al. [55] studied Bi, Na, and Ag segregation at the Cu $\Sigma 5(310)$ boundary, and found that Ag exhibited a segregation energy of ~ 0.8 eV, also affecting cleavage work and interfacial strength. More recently, S.N. Zhevnenko et al. [56] used DFT to evaluate Ni, Co, and Fe segregation in Ag, showing that these elements display pronounced “anti-segregation” behavior (i.e. they do not readily segregate to Ag surfaces/interfaces). Together, these examples suggest that segregation or doping energies in the range of ~ 0.5 – 0.8 eV represent a reasonable reference window for simple metallic systems. Consistent with this, our results for metallic phases Ag (Fig. 4d) and Cu (Fig. 4e) fall within this range, where values below 0.8 eV can generally be regarded as acceptable. Such values indicate good solubility and structural compatibility, implying that Ag and Cu, as the matrix metals in the AgCuSnTi filler alloy, can readily accommodate other filler elements. This promotes the formation of thermodynamically stable solid solutions without substantial energetic penalties.



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Fig. 4. Doping formation energies of brazing phases with different filler elements and doping concentrations: a) TiC, b) CuTi, c) CuSn₃Ti₅, d) Ag, and e) Cu. f) Average doping formation energy for each brazing phase, reflecting the tendency of dopant incorporation and the thermodynamic stability of each phase under multi-element interactions.

IMC such as CuTi (Fig. 4b) exhibit moderate dopability, particularly when dopants substitute at Cu sites, with doping formation energies typically remaining below approximately 1.0 eV. Notably, the CuTi phase displays a distinct asymmetry in doping behavior: substitutions at Cu sites consistently yield lower formation energies, while substitutions at Ti sites result in

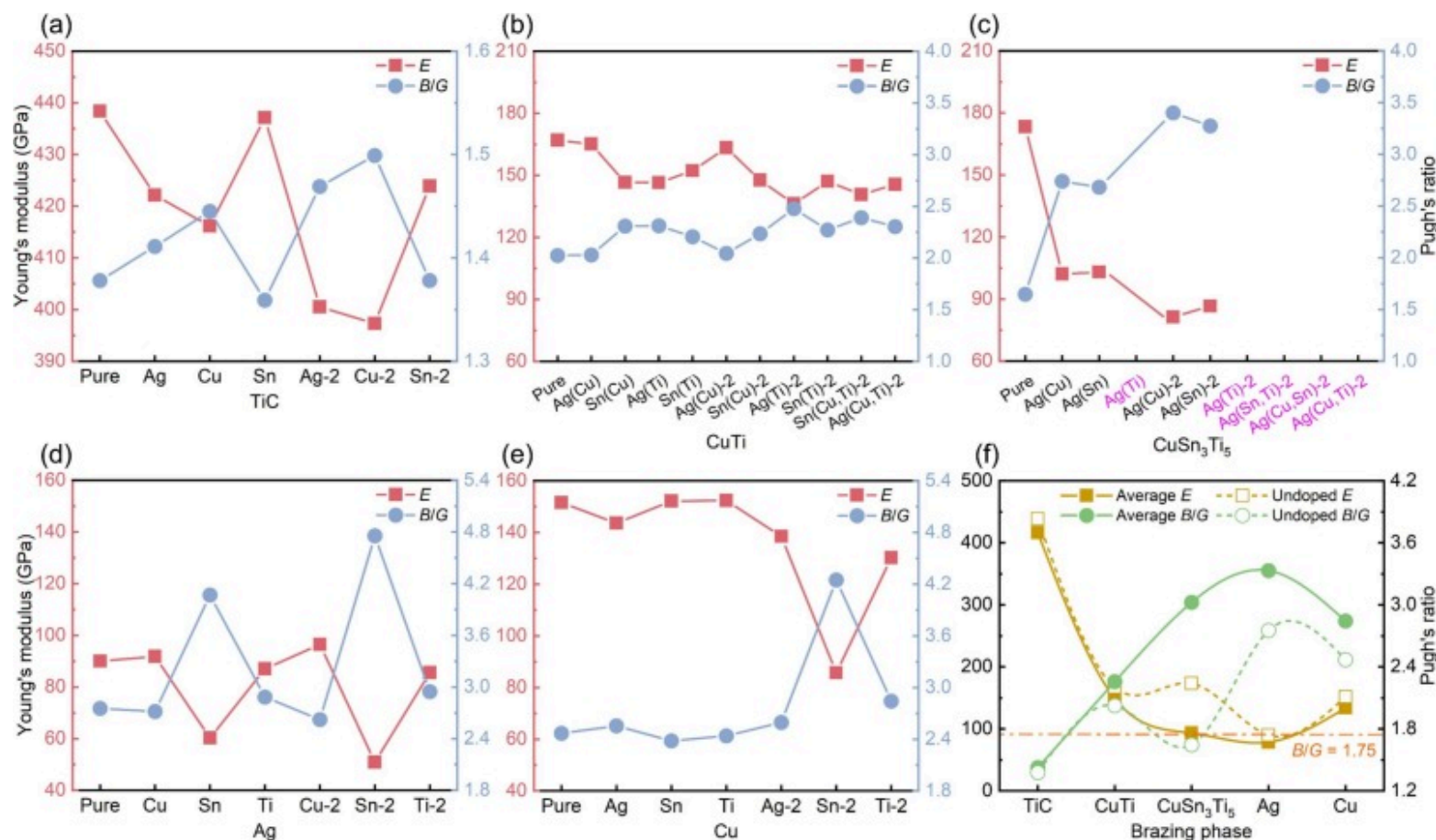
significantly higher energetic costs. This trend likely arises from the higher chemical reactivity and bonding flexibility of Cu atoms compared to the more tightly bound Ti atoms within the CuTi lattice. As a result, Cu sites are thermodynamically more favorable for dopant incorporation. In contrast, strongly bonded phases such as TiC (Fig. 4a) and CuSn₃Ti₅ (Fig. 4c) exhibit significantly higher doping formation energies across all investigated elements and concentrations, often exceeding 8.0 eV. These results indicate poor solubility and strong resistance to lattice substitution, suggesting that these ceramic-like or complex intermetallic phases are thermodynamically unfavorable for dopant incorporation. Attempts to introduce dopants into such rigid lattices may lead to local structural distortions or even second-phase precipitation. Under the multi-element conditions characteristic of the AgCuSnTi filler system, this thermodynamic behavior implies that once these reaction phases form at the interlayer, their compositions are likely to remain close to stoichiometric. Dopants originating from other filler elements, such as Ag, are unlikely to be incorporated into the lattices of these stable interfacial compounds.

To further evaluate the thermodynamic feasibility of dopant incorporation, average doping formation energies were calculated. Lower average values indicate greater structural tolerance and compositional compatibility, as they reflect reduced energetic penalties for substitution. As shown in Fig. 4f, TiC and CuSn₃Ti₅ exhibit the highest average doping energies among all phases, reinforcing their identities as thermodynamically stable and chemically selective compounds. In contrast, Ag and Cu display the lowest average values, confirming their roles as flexible solvent matrices that can accommodate a variety of filler elements with minimal energy cost. This characteristic facilitates microstructural uniformity and enhances wetting adaptability during brazing. Overall, the doping formation energy results align well with experimental observations. Ag and Cu are widely reported to form extensive solid solutions and homogeneous microstructures in AgCu-based filler alloys, whereas CuTi and CuSn₃Ti₅ typically appear as discrete, chemically stable reaction products with limited solubility for additional elements.

3.2.2. Mechanical properties

Understanding the mechanical properties of phases is essential for evaluating the structural reliability of diamond/AgCuSnTi joints. Doping can significantly alter the mechanical response, potentially reducing stiffness or inducing mechanical instability. This is especially important for IMCs, where increased brittleness may promote crack initiation under thermal or mechanical loading, ultimately compromising joint performance. To assess the residual stress accumulation due to thermal expansion mismatch within the reaction layer, Young's modulus (E) was selected as the primary parameter. Given that thermally induced residual stresses are predominantly uniaxial (tensile or compressive), E provides a more direct measure of elastic resistance than bulk modulus (B) or shear modulus (G). Mechanical mismatches between different phases were evaluated by comparing their E

values, which reflect the likelihood of stress concentration at heterogeneous phase boundaries. Additionally, the ductile or brittle character of doped phases was further assessed using the B/G ratio derived from the elastic constants. According to Pugh's criterion, a B/G value above 1.75 indicates ductile behavior, while values below this threshold suggest increased brittleness [57]. To ensure mechanical viability, elastic stability criteria were also applied [58]. These criteria confirm that the strain energy remains positive under small deformations and are essential for ensuring structural integrity. Configurations failing to meet these conditions were classified as mechanically unstable and considered unlikely to form or persist under service conditions. As shown in Fig. 5, doping configurations with pink-colored x-axis labels indicate mechanical instability as determined by these criteria.



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Fig. 5. Mechanical properties of doped brazing phases with different filler elements and doping concentrations: a) TiC, b) CuTi, c) CuSn₃Ti₅, d) Ag, and e) Cu. The undoped (pure) phases are included for comparison. Doping configurations marked in pink on the x-axis are identified as mechanically unstable. f) Average mechanical properties of doped phases versus undoped phases, illustrating the overall influence of filler element interactions on phase stiffness and ductility. For all subfigures with dual y-axes, the left axis represents Young's modulus (E), and the right axis denotes Pugh's ratio (B/G). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The E of doped brazing phases was evaluated using the Hill averages derived from the Voigt-Reuss-Hill (VRH) scheme [59]. Fig. 5a–e presents the mechanical properties of both undoped and doped brazing phases. In all cases, doped supercells exhibit a clear reduction in E compared to their undoped counterparts, indicating a general softening effect induced by elemental substitution. This trend can be attributed to local lattice distortion and bonding heterogeneity introduced by the dopant atoms, which disrupt the regular atomic framework and reduce the efficiency of load transfer within the crystal. Even dopants with intrinsically high stiffness, such as Ti, result in a degree of softening. The softening effect is most pronounced with Sn doping, as Sn possesses inherently lower elastic moduli than Ag, Cu, or Ti. Its incorporation weakens the host lattice through a combination of structural distortion and electronic disruption.

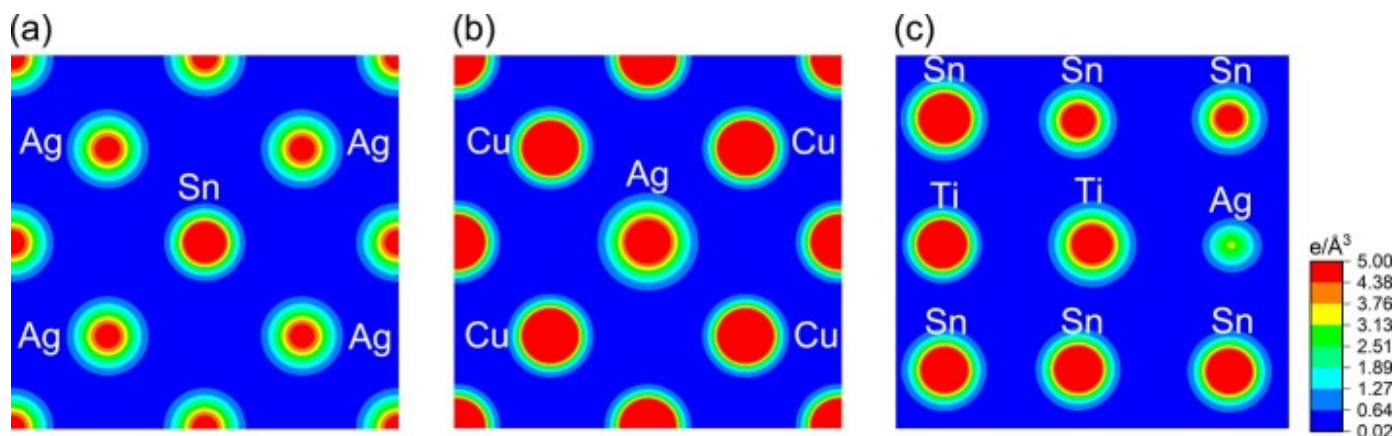
It is worth noting that in Fig. 5d a abnormal phenomenon is observed: substitution of Ag by 1 or 2 Cu atoms leads to a slight increase in E . Unlike Sn or Ti, which tend to introduce significant lattice distortion and weaken metallic bonding in Ag, Cu is chemically and structurally very similar to Ag (both are FCC metals with close atomic radii), but possesses a higher intrinsic stiffness (bulk Cu has a larger Young's modulus than bulk Ag). Therefore, when Cu atoms substitute Ag sites, the resulting solid solution inherits part of Cu's stronger metallic bonding, leading to a modest increase in elastic stiffness. In this case, the electronic and structural compatibility between Ag and Cu minimizes lattice distortion while enhancing bond strength. In summary, the elastic properties of brazing phases are highly sensitive to the intrinsic mechanical characteristics of dopant elements and the atomic-scale structural perturbations they induce. The results further reveal that, in multicomponent alloy brazing systems, the interactions among different elements and their structural compatibility within various brazing phases significantly affect the mechanical performance and service stability of the joint. Therefore, in the design and compositional optimization of filler alloys, it is essential to consider the chemical compatibility and structural tolerance among constituent elements. By rationally tuning the elemental combinations and doping behavior, the robustness of interfacial structures and the overall reliability of the brazed joint can be effectively enhanced.

An interesting phenomenon emerges from Ag-doped CuSn_3Ti_5 configurations, particularly those involving substitution at Ti sites, including Ag(Ti), Ag(Ti)-2, Ag(Cu,Ti)-2, Ag(Sn,Ti)-2, and Ag(Cu,Sn)-2. Ag is a relatively large and chemically soft element with low bonding affinity toward Ti and Sn. When substituted into CuSn_3Ti_5 , it can induce substantial lattice strain and electronic imbalance. Despite the high doping formation energies associated with these configurations, typically around 10eV, which indicate strong thermodynamic instability, structural optimizations were found to converge successfully. This suggests that these configurations may exist as metastable states. However, elastic stability analysis, as shown in Fig. 5c, reveals that these configurations fail to satisfy mechanical stability criteria. This indicates that although the atomic structures appear relaxed and locally stable, they are dynamically unstable and susceptible to spontaneous distortion under mechanical perturbations. This implies that Ag-doped CuSn_3Ti_5 phases may transiently form during high-temperature diffusion but are unlikely to persist upon cooling. This observation offers a possible explanation for the experimentally reported absence of Ag elements in CuSn_3Ti_5 precipitates [22].

To evaluate the collective mechanical response of the brazing system under multi-element doping, the average elastic properties of doped phases were compared with those of their undoped counterparts. As shown in Fig. 5f, the average mechanical properties of the doped brazing phases (represented by hollow markers) demonstrate a clear trend of reduced stiffness and enhanced ductility relative to their undoped counterparts (solid markers). Specifically, Young's modulus (E) is consistently lower across all doped phases, while the corresponding Pugh's ratios (B/G) generally increase. These trends suggest that multi-element interactions in the AgCuSnTi brazing system lead to phase softening and enhanced toughness, which are expected to play a beneficial role in improving joint performance. The reduction in E suggests decreased stiffness, helping to mitigate stress concentrations at brittle-phase interfaces, while the increase in B/G reflects greater plastic deformability, thereby facilitating strain accommodation across heterogeneous-phase boundaries. Importantly, in addition to lowering E values, doping leads to a more continuous distribution of stiffness among the various phases, in contrast to the sharp mechanical discontinuities observed in the undoped system. This smoother mechanical gradient is expected to reduce local residual stress concentrations and improve mechanical compatibility. Overall, these findings underscore the significance of tailoring filler element composition not only for chemical reactivity and wetting, but also for optimizing mechanical compatibility across the microstructure.

Building on the preceding thermodynamic and mechanical analyses, charge density analysis was conducted to further investigate the electronic interactions induced by elemental doping. Three representative doped configurations were selected: Sn-substituted Ag (Sn@Ag), Ag-substituted Cu (Ag@Cu), and Ag-substituted Ti site in CuSn_3Ti_5 (Ag(Ti)@ CuSn_3Ti_5). These cases capture distinct doping behaviors and their implications. Sn@Ag was chosen for its lowest doping formation energy and

pronounced changes in mechanical properties. Ag@Cu reflects the most common elemental combination in AgCu-based fillers, relevant to solid solution formation and wetting behavior. Ag(Ti)@CuSn₃Ti₅ represents a mechanically unstable configuration: despite structural convergence, its elastic instability suggests limited practical applicability. As shown in Fig. 6, the charge distribution around the dopant sites reveals the extent of charge transfer and bonding characteristics induced by filler elements.



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Fig. 6. Charge density maps of representative doped brazing phases: a) Sn-substituted Ag (Sn@Ag), b) Ag-substituted Cu (Ag@Cu), and c) Ag-substituted Ti site in CuSn₃Ti₅ (Ag(Ti)@CuSn₃Ti₅), highlighting the local charge distribution around the dopant filler atoms.

The charge accumulation pattern around the doped Sn atom and its neighboring Ag atoms (Fig. 6a) is comparable to that observed between host Ag atoms, indicating that Sn can be well accommodated within the Ag phase. This suggests that Ag–Sn interactions exhibit metallic bonding characteristics similar to those of Ag–Ag, likely due to their similar electronic configuration and delocalized outer electrons. Sn ($4d^{10}5s^25p^2$) provides a comparable valence shell to Ag ($4d^{10}5s^1$), allowing effective electron delocalization and orbital overlap. The good chemical compatibility also implies that Sn may act as a melting point depressant while remaining incorporated within the reaction-layer phases. Interestingly, Sn substitution at Ag sites leads to a notable decrease in E along with a concurrent increase in ductility indicators (B/G). This mechanical softening may be attributed to the larger atomic radius and lower bond stiffness of Sn compared to Ag, which introduces local lattice distortion and reduces overall stiffness while promoting greater plastic deformability and stress accommodation. However, excessive softening may

compromise the load-bearing capacity, indicating that Sn doping must be carefully optimized to enhance interfacial toughness without sacrificing strength.

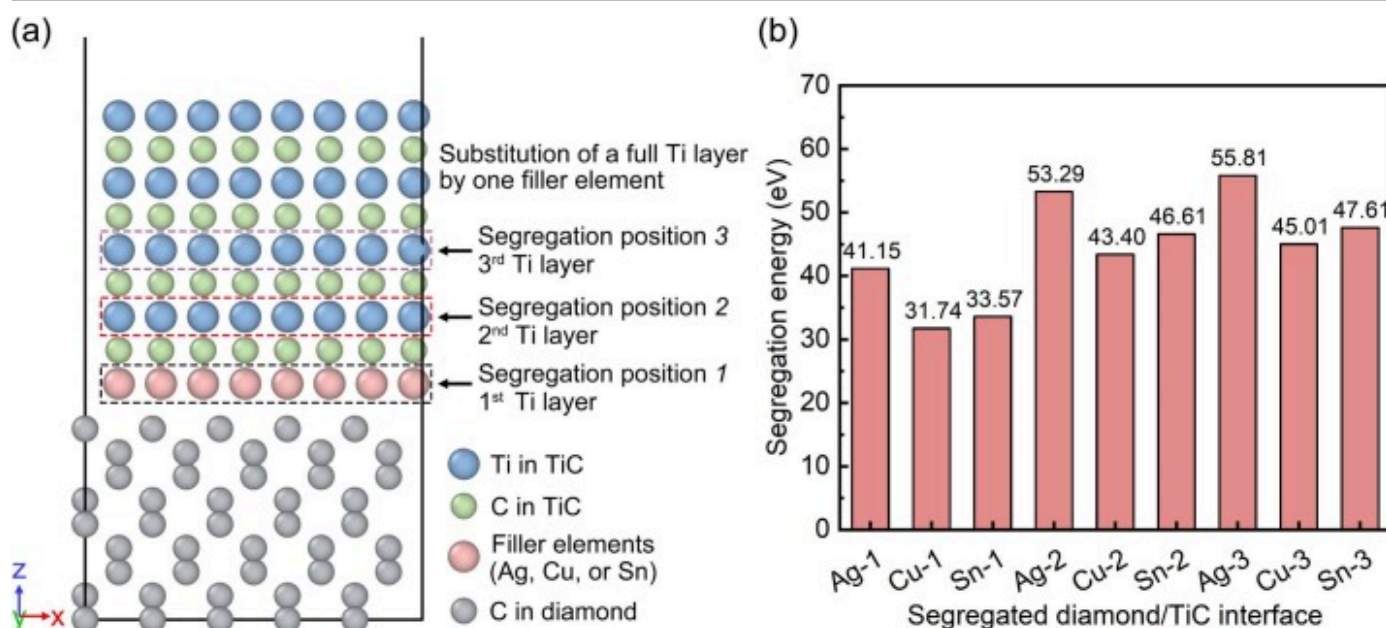
Fig. 6b shows Ag substitution in the Cu phase. The charge density surrounding the Ag dopant closely resembles that of the adjacent Cu atoms. This is consistent with the weaker metallic bonding nature of Ag compared to Cu, which has a higher bonding electron density due to its more compact atomic structure. Nevertheless, the overall similarity in charge distribution confirms the good solubility and lattice compatibility between Ag and Cu. Since Ag and Cu are the dominant constituents in the AgCuSnTi filler alloy, this interaction supports the formation of a stable metallic matrix with minimal electronic or structural disruption, which favors homogeneous microstructures and efficient wetting. In Fig. 6c, the Ag dopant substituted at a Ti site in CuSn_3Ti_5 exhibits a markedly lower charge density than the equivalent Ti atoms. This suggests significantly weaker bonding, likely due to the inability of Ag to form the directional or partially covalent bonds that Ti typically engages in within intermetallic compounds. Ag's filled 4d shell and single 5s valence electron contribute less to the electron density in bonding regions compared to Ti's partially filled $3d^24s^2$ configuration. This electronic disparity likely underlies the mechanical instability of the Ag-substituted CuSn_3Ti_5 configuration, as previously indicated by its elastic constants. This suggests that Ag is unlikely to be stably accommodated within the CuSn_3Ti_5 phase. Such unfavorable elemental interaction may induce local mechanical inhomogeneities in the brazed interlayer, potentially leading to stress concentration.

3.3. Filler element segregation and its influence on adhesion and fracture at the diamond interface

3.3.1. Interfacial adhesion and bonding strength

The strength of brazed joints is governed by the synergistic effects of residual stress and interfacial bonding. Residual stress primarily originates from thermal expansion mismatch among constituent phases and temperature gradients during cooling. When residual stress is minimized through joint design or controlled cooling, interfacial bonding becomes the dominant factor affecting joint integrity. As fracture in diamond-brazed joints typically initiates at the outermost diamond interface, the bonding between diamond and the in-situ formed TiC layer is of particular importance. During TiC formation, additional filler elements may segregate at the interface, modifying local bonding characteristics. Therefore, quantifying the effects of elemental segregation on interfacial strength is critical for understanding elemental inhomogeneity and predicting failure mechanisms.

Building on previous investigations of equilibrium stacking configurations for the diamond (001)/TiC (111) interface (TiC along $x||[1\bar{1}0]$, $y||[11\bar{2}]$, and $z||[111]$, diamond along $x||[1\bar{1}0]$, $y||[110]$, and $z||[001]$), the Ti-terminated hollow-site stacking was identified as the most energetically favorable arrangement [42]. Based on this, interface models were constructed by substituting Ti atoms in the first, second, or third Ti layers of the TiC(111) slab near the interface with Ag, Cu, or Sn atoms, reflecting the alternating Ti-C stacking along the TiC[111] directions. Given the large atomic radii and metallic bonding character of filler elements, substitution at C sites would induce unrealistic lattice distortions. Therefore, Ti-site substitution was employed to simulate segregation in a physically and chemically reasonable manner. To better represent experimentally observed interfacial enrichment, entire atomic layers were substituted instead of isolated atoms. A simplified naming convention is used in the format “*element-layer*”, where, for example, “Ag-1” denotes Ag substitution in the first Ti layer adjacent to the interface. The segregation strategy and substitution sites are schematically illustrated in Fig. 7a.



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Fig. 7. a) Schematic illustration of the segregation modeling strategy and different segregation configurations at the diamond(001)/TiC(111) interface. b) Calculated segregation energies of diamond/TiC interfaces with different filler elements and segregation positions.

To quantitatively assess the thermodynamic driving force for filler element segregation at the diamond/TiC interface, the segregation energy (E_{seg}) was calculated:

$$E_{\text{seg}} = E_{\text{interface}}^{\text{segregated}} - E_{\text{interface}}^{\text{clean}} + \sum_{i=1}^n n_i \mu_i^{\text{host}} - \sum_{j=1}^n n_j \mu_j^{\text{dopant}} \quad (3)$$

where $E_{\text{interface}}^{\text{segregated}}$ and $E_{\text{interface}}^{\text{clean}}$ correspond to the total energies of interfaces with filler element segregation and without segregation (i.e. clean interface), respectively. A negative E_{seg} indicates that the filler element has a thermodynamic preference to occupy the interfacial region, suggesting a tendency for interfacial enrichment. The more negative the E_{seg} , the stronger the segregation tendency, which in turn can significantly impact interfacial chemistry and bonding strength. As shown in Fig. 7b, all calculated E_{seg} for Ag, Cu, and Sn doped at the first, second, and third Ti layers are positive, indicating that these filler elements are thermodynamically less stable at the diamond/TiC interface compared to the bulk TiC region. This trend contrasts with typical grain boundary segregation behavior, where segregation energy is often negative due to lower coordination or structural disorder at boundaries that favor impurity incorporation. In the case of the diamond/TiC interface, however, the interface is well-ordered and structurally coherent, especially with the Ti-terminated hollow-site stacking, which results in limited available sites and stronger local bonding, making segregation energetically unfavorable. Among the three elements, Ag consistently exhibits the highest segregation energy across all substitutional positions. Such behavior implies that Ag is unlikely to segregate directly at the diamond/TiC interface and may instead remain in the matrix or enrich at other less ordered regions, such as grain boundaries or triple junctions. This limited interfacial solubility of Ag could lead to phase separation or microstructural discontinuity near the interface.

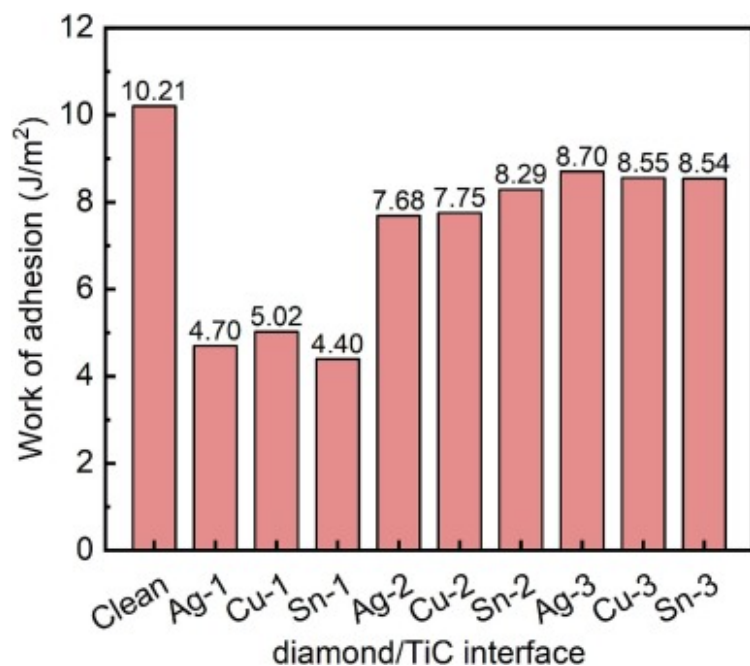
Furthermore, a clear trend is observed where E_{seg} increases with increasing distance from the interface (i.e. from position 1 to 3). This suggests that segregation becomes progressively less favorable deeper into the TiC lattice, highlighting a weak driving force for long-range diffusion toward the interface. From the perspective of interface adhesion, such segregation resistance may help maintain the chemical and structural integrity of the TiC reaction layer, which is critical for stable bonding with diamond. However, for elements like Cu or Sn that exhibit moderately lower segregation energies near the interface, local enrichment could still occur under kinetic conditions, potentially tuning interfacial bonding characteristics.

To further evaluate the impact of elemental segregation on interfacial performance, the work of adhesion (W_{ad}) was introduced as a quantitative measure of interfacial bonding strength. W_{ad} represents the energy required to separate the interface into two

fully relaxed slabs, which can be calculated as:

$$W_{\text{ad}} = \frac{1}{A} (E_{\text{slab-1}} + E_{\text{slab-2}} - E_{\text{interface}}) \quad (4)$$

where $E_{\text{slab-1}}$ and $E_{\text{slab-2}}$ are the total energies of two isolated slabs, respectively, $E_{\text{interface}}$ is the total energy of the relaxed interface model (consisting of slab-1 and slab-2), and A is the interfacial area. An increase in W_{ad} indicates enhanced interfacial bonding due to segregation, whereas a decrease suggests weakened adhesion. As shown in Fig. 8, the W_{ad} for all segregated diamond/TiC interface configurations is lower than that of the clean interface (10.21 J/m^2), indicating that elemental segregation consistently weakens interfacial bonding. This suggests that the presence of Ag, Cu, or Sn at or near the interface disrupts the optimal chemical bonding environment, thereby reducing the energetic stability of the interface. Such degradation in adhesion strength may contribute to interfacial debonding or crack initiation, which has been frequently observed in experiments and is a critical factor affecting the mechanical reliability of brazed joints. Since segregation is thermodynamically or kinetically inevitable during high-temperature joining, understanding its impact is essential for accurate evaluation of joint performance.



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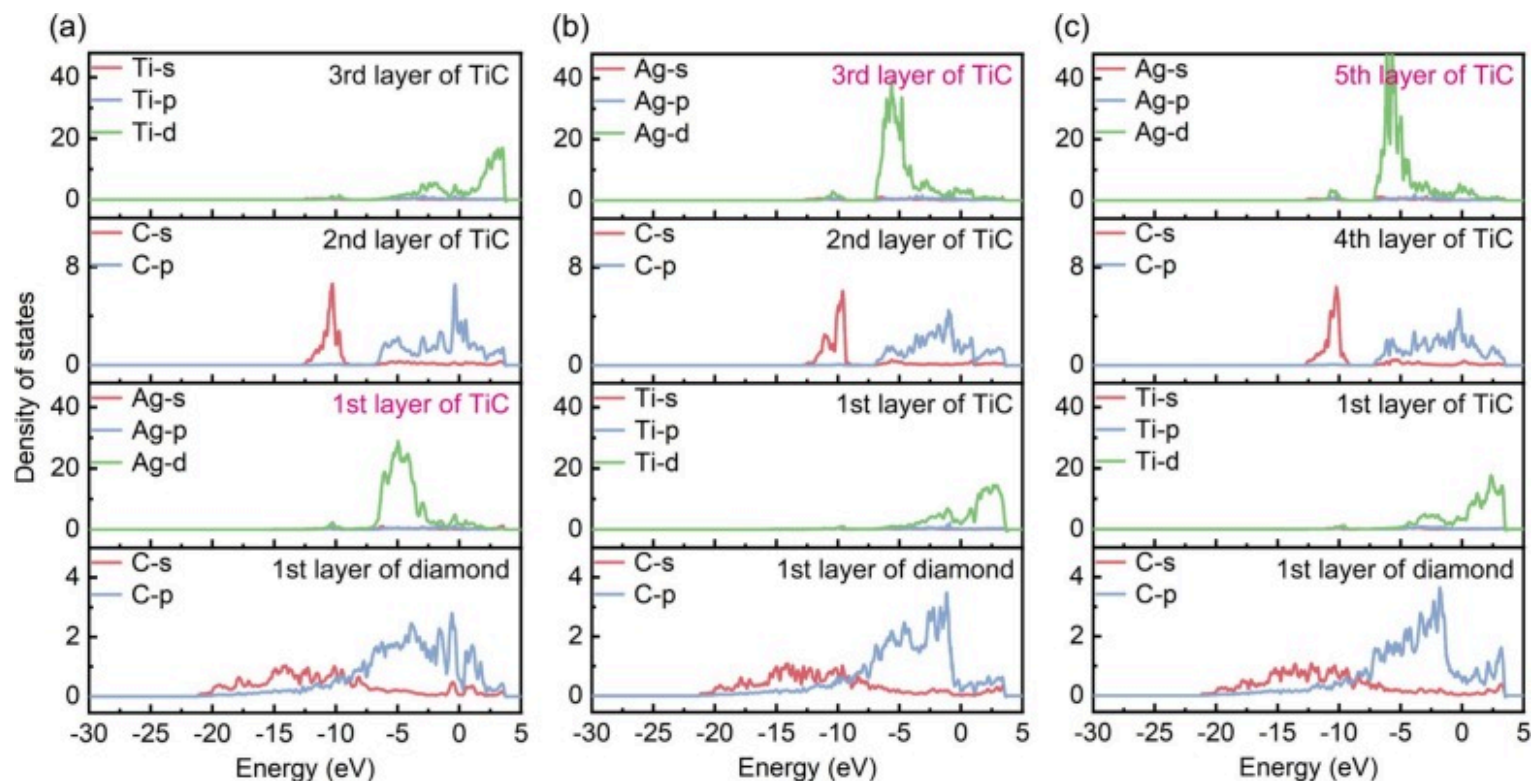
Fig. 8. Work of adhesion for segregated diamond/TiC interfaces with different filler elements and segregation positions. The Clean interface is included for comparison.

Furthermore, a clear positional trend is observed: as the segregated layer moves farther from the interface (from position 1 to position 3), its influence on W_{ad} gradually diminishes. This behavior can be attributed to the reduced electronic and structural perturbation imposed on the bonding plane when dopant atoms are situated deeper within the bulk region. Atoms located at or near the interface directly affect charge redistribution and orbital overlap across the interface, thereby exerting a stronger impact on interfacial bonding. Interestingly, although Ag exhibits the highest segregation energy among the three elements, indicating it is the least likely to segregate to the interface, it does not lead to the lowest W_{ad} . Instead, the Sn-1 configuration results in the weakest interface (4.40J/m^2). This suggests that, despite Ag introducing substantial lattice strain and being thermodynamically unfavorable for segregation, it may still retain some degree of interfacial bonding due to residual orbital overlap or limited disruption of the interfacial electronic environment. In contrast, Sn, which has the highest electronegativity, appears to weaken interfacial bonding more effectively. This may result from its ability to alter local bonding characteristics without contributing significantly to new bonding interactions. These findings highlight the complex interplay between segregation thermodynamics, local electronic structure, and interfacial adhesion strength. They underscore the importance of evaluating both the likelihood of segregation and the extent of bonding disruption when assessing the role of filler elements in determining joint integrity.

To further elucidate how elemental segregation alters the local bonding environment, projected density of states (PDOS) analyses were conducted. PDOS reveals orbital-level contributions and the electronic states near the Fermi level (E_F), which are critical to interfacial bonding strength and stability. Layer-resolved PDOS was extracted for three Ag-segregated configurations, Ag-1 (near-interface), Ag-2 (sub-interface), and Ag-3 (far-interface), to assess how the segregation depth affects both the original diamond/TiC bonding and the possible formation of new chemical bonds. TiC-side atomic sequences near the interface for each configuration, counted from the diamond surface are as follows: Ag-1 (Ag-C-Ti-C-Ti-...), Ag-2 (Ti-C-Ag-C-Ti-...), and Ag-3 (Ti-C-Ti-C-Ag-C-Ti-...). For clarity, the PDOS analysis focused on selected layers: the outermost diamond-side carbon, the TiC layers bonded to diamond, the Ag-substituted layer, and the carbon layer adjacent to Ag. These correspond to C/Ag-C-Ti-C-Ti-... for Ag-1, C/Ti-C-Ag-C-Ti-... for Ag-2, and C/Ti-C-Ti-C-Ag-C-Ti-... for Ag-3, with the analyzed layers highlighted in bold.

The PDOS of the segregated Ag atoms shows that their electronic contributions are dominated by Ag-*d* orbitals, whose interactions with surrounding states vary significantly with segregation depth. In the Ag-1 configuration (Fig. 9a), where Ag substitutes the first Ti layer directly at the interface, the PDOS profile of the diamond surface reveals a reduction in hybridization

peak intensity compared to deeper segregation cases. Specifically, the Ag-*d* states display minimal overlap with the *s* and *p* states of nearby diamond C atoms, with only narrow resonances appearing just below the E_F . Furthermore, the PDOS intensity of the interfacial C atoms is substantially diminished, indicating a general weakening of orbital interactions across the interface. Additional states emerging above E_F may correspond to unoccupied antibonding states or charge redistribution effects caused by Ag incorporation, both indicative of a destabilized bonding environment. These features suggest that Ag-C interactions are weak and lack the directional covalency seen in strong bonds, likely due to poor orbital hybridization between Ag-*d* and C-*p* states. The limited charge overlap and mismatch in electronegativity and orbital symmetry between Ag and C further hinder effective bond formation. As a result, the original Ti-C bonding network is disrupted, giving rise to a loosely bonded interface and a marked decrease in adhesion strength. These electronic insights align with the reduced W_{ad} observed for the Ag-1 configuration, confirming that Ag segregation at the interface leads to weakened bonding and compromised mechanical integrity of the diamond interface.



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Fig. 9. PDOS of diamond/TiC interfaces under different Ag segregation configurations: a) Ag-1, b) Ag-2, and c) Ag-3. The specific segregation layer is indicated in pink at the upper-right corner of each subplot, and the Fermi level is set to zero. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

In contrast, the Ag-2 configuration (Fig. 9b), where Ag substitutes the second Ti layer (also the third atomic layer of TiC), preserves the original Ti-C bonding network at the interface. The PDOS shows typical Ti-*d* and C-*p* hybridization features near E_F , indicating sustained interfacial cohesion. Although the Ag-*d* orbitals in the sub-interface position still exhibit moderate hybridization with neighboring C-*p* states, their impact on the interfacial electronic structure is limited. As a result, the Ag-2 configuration maintains stronger adhesion than Ag-1, owing to less disruption of the primary bonding layer. In the Ag-3 case (Fig. 9c), where Ag is located deeper within the TiC slab (third Ti layer or fifth atomic layer overall), the PDOS profiles approach those of bulk TiC. While Ag-*d* states continue to show minor hybridization with surrounding C-*p* orbitals, their influence on the interfacial bonding becomes negligible. Notably, the PDOS peaks of interfacial Ti atoms become sharper and more intense compared to those in Ag-2, indicating enhanced orbital hybridization and a relatively undisturbed bonding environment. This explains the observed recovery in W_{ad} for Ag-3, as the electronic effects of segregation are spatially decoupled from the interface region.

The PDOS analysis reveals configuration-dependent changes in orbital hybridization at the diamond/filler interface. These variations might be experimentally accessed. 1) Valence-band XPS can be semi-quantitatively reconstructed from PDOS, allowing direct comparison of hybridization strength [60,61]. 2) ELNES/EELS probes the unoccupied DOS, where reduced pre-peak intensity at edges indicates weakened *p-d* hybridization and interfacial bonding [62]. 3) Phase-selective XRD/EDS or fracture-surface features indicative of Ti-C reaction layers versus filler element-rich weakly bonded regions. These correlations provide a useful framework linking electronic structure to phase constitution and joint reliability.

3.3.2. Tensile fracture simulation

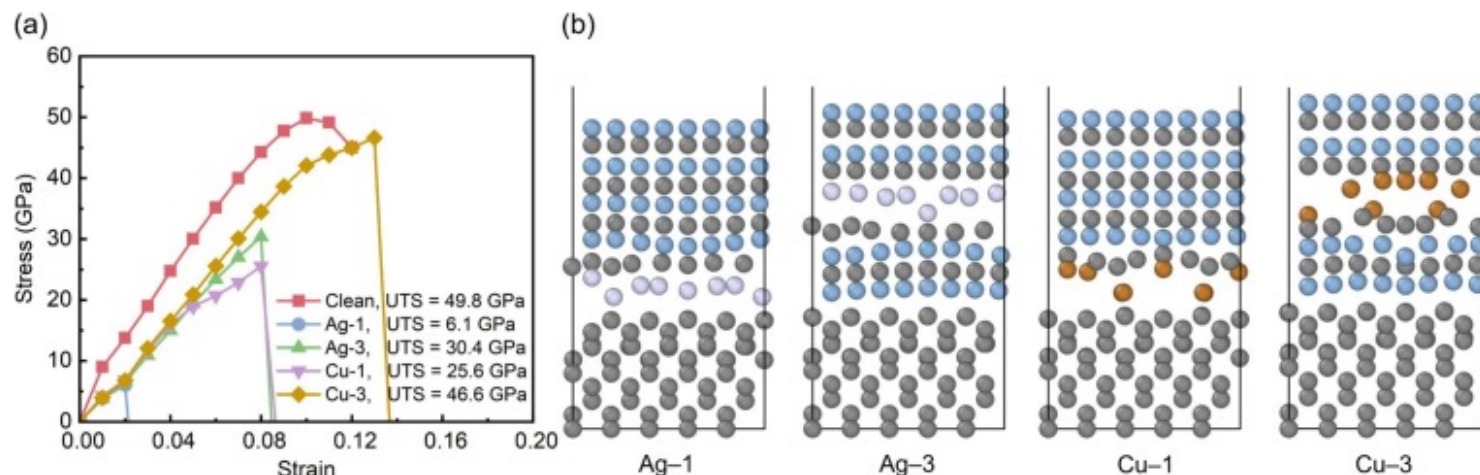
Following the previous section, where representative segregation configurations were analyzed to uncover the electronic origins of bonding variation, it was found that near-interface Ag segregation can significantly weaken the interface due to disrupted orbital hybridization. To further validate these findings from a mechanical perspective, it is necessary to evaluate the dynamic fracture behavior of the diamond/TiC interface under loading conditions. The fracture location, whether it occurs at the diamond/TiC interface, within the TiC, or in the segregation-enriched sites, provides details into the critical bonding sites and

mechanical vulnerability introduced by specific filler elements. In DFT calculations, stress values were obtained by calculating the derivative of total energy for applied strain, allowing the extraction of ideal fracture strength. The stress (σ) corresponding to an applied strain (ε) is given by [63]:

$$\sigma = \frac{1}{V_0} \frac{dE}{d\varepsilon} \quad (5)$$

where V_0 is the equilibrium volume of the unstrained unit cell (before deformation). E is the total energy of the interfaces at a given strain of ε . A uniaxial strain of 0.01 was applied incrementally along the z -direction, with atomic positions fully relaxed at each step, except for the top and bottom three atomic layers, which were fixed to simulate boundary conditions. Four representative segregation configurations: Ag-1, Ag-3, Cu-1, and Cu-3, where Ag or Cu atoms substitute Ti atoms in the first (interfacial) or third (bulk-like) Ti layer of TiC. Ag and Cu were selected as representative segregation elements for interfacial simulations due to their dominant concentrations in AgCuSnTi filler alloys and contrasting chemical characteristics. Ag, with its weak bonding affinity and large atomic radius, represents segregation-induced interfacial degradation, while Cu serves as a transitional species with moderate bonding potential and structural adaptability at the diamond/TiC interface.

Fig. 10a shows the tensile stress-strain responses for clean and segregated diamond/TiC interfaces. The peak of each tensile stress-strain curve is defined as the ultimate tensile strength (UTS), representing the maximum stress that the interface can sustain before fracture occurs. For the clean interface, stress decreases gradually after reaching UTS, which reflects a progressive debonding process. In contrast, all segregated interfaces exhibit an abrupt stress drop immediately after UTS, rapidly falling into negative values. According to Eq. 5, this negative stress means that further strain leads to spontaneous energy release, confirming a mechanically unstable state and instantaneous brittle fracture. Notably, the UTS of near-interface segregated configurations (Ag-1 and Cu-1) is much lower than that of the clean interface, declaring their severe weakening effect and susceptibility to interfacial decohesion.



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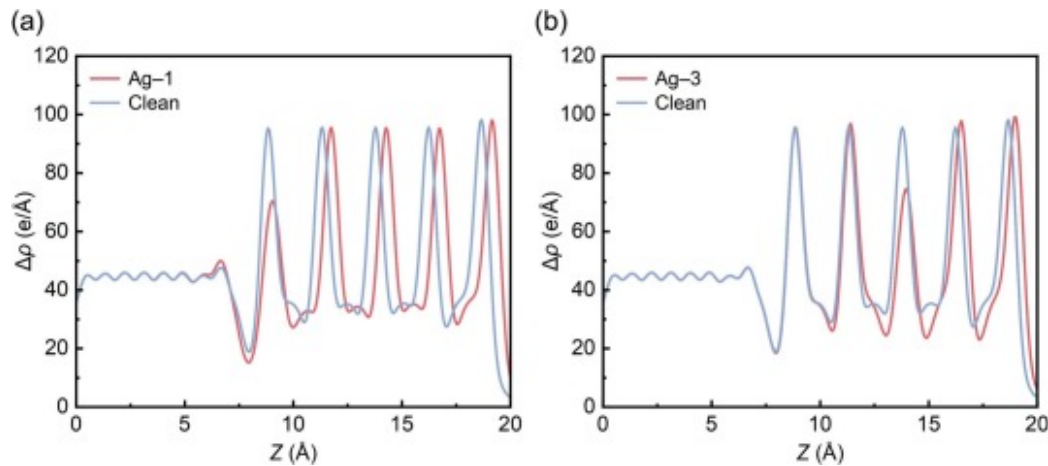
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Fig. 10. a) Tensile stress-strain curves of the diamond/TiC interface under typical segregation configurations, with the clean interface included for comparison. b) Atomic arrangements at UTS, illustrating localized fracture characteristics.

Fig. 10b presents the atomic arrangements of the interfaces at UTS. For the segregation configurations located nearest to the interface (Ag-1 and Cu-1), fracture occurs predominantly along the original diamond/TiC boundary, with some residual atoms still adhering to the diamond surface. This indicates a failure mechanism governed by direct weakening of interfacial bonding. In contrast, for configurations with segregation deeper within the TiC side (Ag-3 and Cu-3), the fracture planes propagate directly through the segregated zone. This suggests that such regions accumulate higher residual stresses due to severe lattice/electronic mismatch, effectively becoming preferential crack initiation sites. Therefore, large-scale elemental enrichment in the TiC layer, as sometimes revealed by SEM/EDS mapping in real joints, is detrimental to joint reliability and should be avoided. The difference in fracture paths implies that nearest-interface segregation primarily disrupts interfacial adhesion, whereas deeper segregation introduces internal planes of weakness within the reaction layer. In summary, chemical inhomogeneity can lead to unpredictable fracture either along the interface or within the interlayer.

To further elucidate the mechanisms behind fracture modes and to understand how segregation affects bonding strength at the electronic level, the planar-averaged charge density along the vertical interface was plotted, which reflects the extent of electronic overlap between adjacent atomic planes. Higher charge density generally corresponds to stronger covalent or metallic

bonding, as it indicates greater charge accumulation and orbital hybridization across the interface. As shown in Fig. 11, the Ag-segregated configurations exhibit a pronounced reduction in charge density within the vicinity of the segregated layers compared to the clean interface. Specifically, the charge density minima near the Ag-1 and Ag-3 layers are significantly lower, indicating a depletion of electron density in these critical bonding regions. This suggests that the introduction of Ag weakens the local bonding network by reducing the extent of electronic sharing. Such weakening is consistent with the previously observed decrease in W_{ad} and the diminished hybridization features in the PDOS. These combined results confirm that the regions with the lowest interfacial charge density effectively constitute mechanically vulnerable zones, where cracks preferentially nucleate along the segregated interfaces.



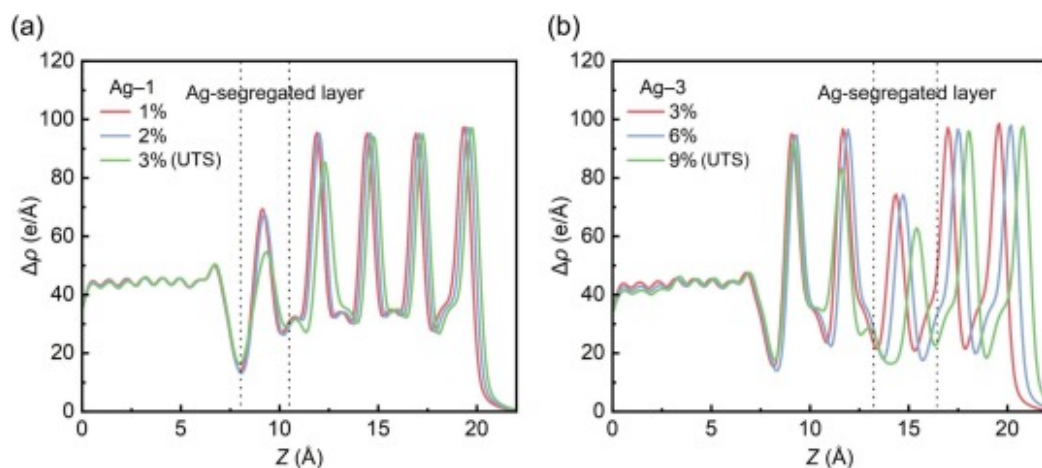
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Fig. 11. Planar-averaged charge density profiles of the equilibrium (unstrained) diamond/TiC interfaces under different Ag segregation configurations: a) Ag-1, and b) Ag-3. The clean interface is included for comparison.

Fig. 12 shows the planar-averaged charge density of Ag-segregated diamond/TiC interfaces under different applied strains. It can be observed that the charge density in the diamond region remains nearly unchanged across all strain levels, indicating that the diamond side maintains an undeformed state with negligible electronic redistribution during loading. In contrast, most charge density variations occur within the reaction layer on the diamond surface. Notably, at the UTS state, the charge density in the segregation layer decreases sharply, indicating a significant loss of electron accumulation and bonding strength at this location.

This sharp drop is consistent with the onset of fracture and corresponds to the maximum tensile load the interface can sustain. Interestingly, for the Ag-3 segregation configuration, the TiC atomic layers directly adjacent to the diamond surface show minimal charge density variation even at large strains. This suggests that the interfacial bonding between diamond and the TiC layer was intrinsically stronger and more stable compared to the TiC interior. Consequently, fracture does not initiate at the immediate interface but rather occurs within the deeper TiC region where Ag segregation was located, confirming that the segregated zone far from the interface acts as the mechanical weak point.



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Fig. 12. Planar-averaged charge density profiles of Ag-segregated diamond/TiC interfaces under different tensile strains: a) Ag-1, and b) Ag-3.

Simulation results indicate that elemental segregation in the atomic layer closest to the diamond interface causes the most pronounced reduction in interfacial bonding strength and triggers fracture initiation at the earliest stage during tensile loading. This highlights the high sensitivity of interfacial adhesion to local chemical inhomogeneity, underscoring the critical importance of controlling filler element distribution. In particular, preventing the excessive segregation of low-bonding-energy elements such as Ag and Sn at the immediate interface is essential for maintaining joint reliability. To mitigate these issues, precise compositional design and process optimization, including adjustments to heating rate, holding time, and filler formulation, are

required. Tailoring interfacial chemistry through strategic alloying or surface modification also presents a promising pathway to enhance the bonding strength of diamond brazed joints.

Importantly, the predicted segregation behaviors and mechanical gradients may be validated experimentally. For instance, nanoindentation of individual reaction layers can provide quantitative information on local stiffness variations, while high-resolution TEM/EDS mapping can directly capture elemental segregation at the nanoscale. In addition, in-situ micromechanical testing of thin reaction layers could further correlate microstructural features with mechanical responses. Such combined approaches would not only help to verify the theoretical predictions but also strengthen the microstructure-property understanding of diamond brazed joints.

4. Conclusions

Given the widespread application of AgCuSnTi filler alloys in diamond brazing, this study investigated the phase constitution and interfacial adhesion of the reaction layer, with particular emphasis on the multi-element interactions and segregation effects.

- 1) The adsorption behaviors of Ag, Cu, Sn, and Ti atoms were investigated on three distinct sites above the diamond(001). Ti and Sn exhibited the highest adsorption energies, with Ti demonstrating a pronounced preference across most configurations. Charge density differences confirmed Ti's role as a dominant wetting agent, enhancing interfacial bonding by forming robust electronic interactions with surface carbon atoms.
- 2) To reflect the compositional complexity of the interlayer, key reaction phases (TiC, CuTi, CuSn₃Ti₅, Ag, and Cu) were doped with additional filler elements. The calculated doping formation energies and mechanical parameters indicated distinct variations in thermodynamic stability and stiffness/ductility. Several doping configurations were found to degrade phase stability. Notably, mechanical gradients across constituent phases were expected to influence residual stress and phase compatibility. Charge density maps clarified how filler-host interactions redistribute electronic charge, affecting bonding characteristics and potentially altering mechanical performance.
- 3) A layer-by-layer segregation modeling strategy was employed for the diamond/TiC interface to explore the effects of elemental segregation. Segregation energies and work of adhesion identified configurations that

impair interfacial bonding, with the most severe weakening observed when segregation occurred in the atomic layer closest to the interface. Layered PDOS revealed that reduced orbital hybridization and peak intensity correlated with lower adhesion strength. First-principles tensile simulations further captured fracture initiation and ideal strength, consistent with static work of adhesion. Charge density maps illustrated the evolution of bonding interactions during tensile loading.

- 4) Ti acted as the primary active element, forming strong Ti–C bonds and a thin TiC layer that enhanced adhesion and wetting. Cu served as a supporting matrix, smoothing the mechanical gradient and improving strain accommodation. Ag provided matrix fluidity, but interacts weakly with C, its enrichment at the interface should be avoided. Sn lowered the liquidus and improved spreadability, but excess at the interface disrupted Ti–C bonding and promoted brittle regions. Moreover, Sn incorporation into brazing reaction phases was found to cause the most significant changes in mechanical properties, indicating that its aggregation must be strictly controlled. In particular, Sn-rich regions revealed by elemental mapping might act as stress concentration sites.

CRedit authorship contribution statement

Jiahe Zhou: Writing – original draft, Methodology. **Xiongying Jiang:** Investigation. **Zhi Xiang:** Investigation. **Hao Wu:** Investigation. **Yuyang Li:** Investigation. **Chuanyang Lu:** Writing – review & editing. **Huaxin Li:** Investigation. **Yuwen Cheng:** Methodology, Investigation. **Jianguo Yang:** Funding acquisition, Conceptualization. **Yanming He:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Recommended articles

Data availability

Data will be made available on request.

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